

SunHyuk Hwang

✉ ericwhale0@gmail.com | 🌐 sunhyuk.dev | 🐙 [GitHub](#) | 🎓 [Google Scholar](#) | 🔗 [LinkedIn](#)

RESEARCH INTERESTS

I work on the security of satellite systems, with a current focus on the RF and communication-protocol layer — signal reception, protocol analysis, and exploitation. Satellite security research has lacked a hands-on, reproducible platform, so I built open satellite signal simulators (VSA, MENKAR) that reproduce the satellite environment without specialized hardware or a real satellite. Alongside them I gathered hands-on experience receiving live satellite signals, and I now want to take that into security research on real systems: eavesdropping on LEO satellite downlinks, satellite signal spoofing, satellite system exploitation, and the hardware security of satellite user terminals such as Starlink.

EDUCATION

M.S., Computer Engineering (Convergence Security) — Kyung Hee University Mar 2025 – Feb 2027
(expected)

Seoul, Republic of Korea

- Graduate researcher, PWNLAB cybersecurity research lab (2025 – present).
- Research focus: satellite communication security.
- GPA: 4.17 / 4.3 (in progress).
- Advisor: Prof. Dae-Hee Jang.
- Thesis: *A Study on the Design of a Virtual Satellite Antenna (VSA) System for Satellite RF Hacking Research* (TBD).

B.S., Computer Engineering — Kyung Hee University

Mar 2019 – Feb 2025

Seoul, Republic of Korea

- Undergraduate researcher, PWNLAB cybersecurity research lab (2023 – 2024).
- Research focus: drone security.

PUBLICATIONS

All venues below are Korean domestic; papers are written in Korean.

1. **S. Hwang** et al., “Design and Implementation of a Virtual Satellite Antenna System for Satellite RF Hacking Research,” *Conference on Information Security and Cryptography — Summer (CISC-S’26)*, Busan, Korea, May 2026. Oral presentation.
2. J. Park, **S. Hwang**, and C. Park, “A Practical Tool for Analyzing Security Threats in Satellite Communications,” *Conference on Information Security and Cryptography — Summer (CISC-S’25)*, Jun 2025.
3. **S. Hwang**, D.-H. Jang, and C. Park, “Analysis of Security Threats in Satellite Control Communication Using the NOS3 Satellite System Simulator,” *Review of KIISC (Korea Institute of Information Security & Cryptology)*, vol. 35, no. 2, pp. 5–10, Apr. 2025. [\[paper\]](#)
4. **S. Hwang** et al., “Research on Exploiting Open-Source Ground Stations through Satellite Spoofing Signal,” *Korea Computer Congress (KCC 2024)*, Jun 2024. [\[paper\]](#)
5. **S. Hwang** et al., “Analysis of the Communication Protocol in Open-Source Drone (PX4),” *Conference on Information Security and Cryptography — Winter (CISC-W’23)*, Dec 2023.

RESEARCH EXPERIENCE

Graduate Researcher — Satellite Cybersecurity

Mar 2025 – present

PWNLAB, Kyung Hee University

- Research on the RF and protocol layer of satellite security — operating real and virtualized ground stations to study signal reception, tracking, and exploitation.
- Creator of [OpenVSA](#) and [Mr.Radio](#) (detailed under Selected Projects) — open tools that make satellite RF security research and education hands-on.

Ground System Lead (RLE) — KARI-hosted CubeSat competition

Jun 2026 – present

KHU team · 7th Korean CubeSat Competition (KASA / KARI)

- Lead the ground-system software segment; designing the Mission Operations Center (MOC) that commands the satellite and processes its telemetry and mission data.

- Government-funded defense satellite security research.

TEACHING EXPERIENCE

Teaching Assistant — Information Security (Undergraduate)

Jun 2026

Kyung Hee University · Instructor: Prof. Dae-Hee Jang

- Led two hands-on RF-hacking lab sessions covering RF attack fundamentals and practical signal exploitation.

RF Security Instructor — PWNLAB

Mar 2024 – present

PWNLAB, Kyung Hee University

- Designed and deliver internal RF-security training for lab members, introducing RF attack surfaces and hands-on exploitation techniques.

TALKS & DEMONSTRATIONS

• DEF CON 2026 Aerospace Village

Aug 2026

Co-developed and present hands-on satellite-hacking scenarios built on OpenVSA, in collaboration with IERAE, for visitors at the Aerospace Village, Las Vegas, NV, USA.

SELECTED PROJECTS

MENKAR — Satellite System Virtualization

2026 – present

Desktop application · personal project

- Virtualizes an entire satellite system — signal simulation, RF hardware virtualization, satellite virtualization, and a live link to the ground station on mrradio.kr — so a complete satellite RF link runs end to end, from a virtual spacecraft down to the ground station.

VSA / OpenVSA — Virtual Satellite Antenna

github.com/whal-e3/OpenVSA

- Desktop application that virtualizes a full satellite ground station — GPredict integration for real-time tracking, IQ signal-file analysis, and multi-antenna beam-pattern simulation — enabling hands-on satellite RF security research without physical hardware.

Mr.Radio — RF-Hacking Wargame Platform

mrradio.kr

- Browser-based RF-hacking wargame with virtualized signal environments — reception, demodulation, and decoding without physical SDR hardware. Challenges span general RF hacking up to satellite communication hacking, so players build RF fundamentals and then take on satellite comms. Over 150 registered users.

Hack-the-Drone CTF — RF Challenges & Operations

2023, 2024

hackthedrone.org

- Created challenges — including drone RF hacking — across 2023 and 2024, and helped run the competition, including building the finals event server in 2023.

HONORS & AWARDS

- **AUTOCRYPT Special Award** — Automotive Hacking Competition (AutoHack), EXCO Daegu finals, Nov 2024. Four-person Kyung Hee University team; hands-on CAN-bus vehicle control (door locks, wipers, headlights) and 433.92 MHz key-fob replay and jamming with a HackRF.
- **HackTheon Sejong — Finalist** — international collegiate CTF hosted by Sejong City.

TECHNICAL SKILLS

- **Software-defined radio:** full-stack satellite RF work — from installing and aiming antennas to receive live signals, through SDR hardware (HackRF One, USRP), to analysis software (GNU Radio, Inspectrum) for demodulation and decoding.
- **Satellite systems:** satellite communication protocols, the NOS3 simulator, NASA's core Flight System (cFS), the TinyGS ground-station stack, and the CCSDS standard protocol.
- **Systems hacking:** reverse engineering and binary exploitation (IDA Pro).